Computação Científica para a Indústria · IFSC

Mosquitos pelo som

Detecção de Aedes aegypti
com redes recorrentes



Câmpus Florianópolis · foto institucional

Luiz Eduardo do Nascimento Ciarallo

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Identificar Aedes para orientar a ação

Hoje: vistorias e coleta.
Identificação morfológica com lupa ou microscópio.
Áudio: triagem automática do voo para sinalizar Aedes e apoiar o monitoramento.





Pneus e recipientes com água · SES-RS / SESA-ES

Monitorar Aedes para apoiar a prevenção da dengue.

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Uma entrada de áudio, três respostas

Objetivo do detector:

Aedes aegypti
Outro mosquito
Ruído / sem mosquito



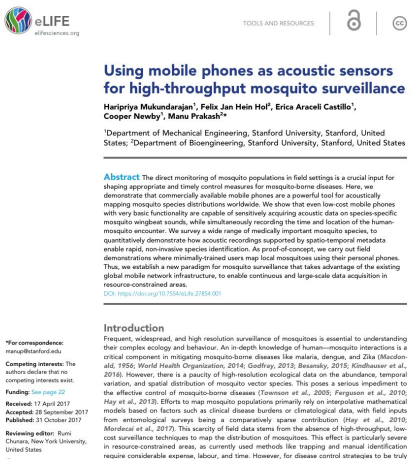
Aedes aegypti · fotografia CDC

Detectar o alvo e separar o ruído.

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Três estudos orientaram a proposta

Mukundarajan · 2017



eLIFE
elifesciences.org

TOOLS AND RESOURCES

Using mobile phones as acoustic sensors for high-throughput mosquito surveillance

Haripriya Mukundarajan[1], Felix Jan Hein Hol[2], Erica Araceli Castillo[1], Cooper Newby[1], Manu Prakash[2]*

[1]Department of Mechanical Engineering, Stanford University, Stanford, United States; [2]Department of Bioengineering, Stanford University, Stanford, United States

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Competing interests: The authors declare that no competing interests exist.

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Reviewing editor: Rumi Chunara, New York University, United States

Abstract The direct monitoring of mosquito populations in field settings is a crucial input for shaping appropriate and timely control measures for mosquito-borne diseases. Here, we demonstrate that commercially available mobile phones are a powerful tool for acoustically mapping mosquito species distributions worldwide. We show that even low-cost mobile phones with very basic functionality are capable of sensitively acquiring acoustic data on species-specific mosquito wingbeat sounds, while simultaneously recording the time and location of the human-mosquito encounter. We survey a wide range of medically important mosquito species, to quantitatively demonstrate how acoustic recordings supported by spatio-temporal metadata enable rapid, non-invasive species identification. As proof-of-concept, we carry out field demonstrations where minimally-trained users map local mosquitoes using their personal phones. Thus, we establish a new paradigm for mosquito surveillance that takes advantage of the existing global mobile network infrastructure, to enable continuous and large-scale data acquisition in resource-constrained areas.

DOI: https://doi.org/10.7554/eLife.27854.001

Introduction

Frequent, widespread, and high resolution surveillance of mosquitoes is essential to understanding their complex ecology and behaviour. An in-depth knowledge of human—mosquito interactions is a critical component in mitigating mosquito-borne diseases like malaria, dengue, and Zika (*Macdonald, 1956*; *World Health Organization, 2014*; *Godfray, 2013*; *Besansky, 2015*; *Kindhauser et al., 2016*). However, there is a paucity of high-resolution ecological data on the abundance, temporal variation, and spatial distribution of mosquito vector species. This poses a serious impediment to the effective control of mosquito-borne diseases (*Townson et al., 2005*; *Ferguson et al., 2010*; *Hay et al., 2013*). Efforts to map mosquito populations primarily rely on interpolative mathematical models based on factors such as clinical disease burdens or climatological data, with field inputs from entomological surveys being a comparatively sparse contribution (*Hay et al., 2010*; *Mordecai et al., 2017*). This scarcity of field data stems from the absence of high-throughput, low-cost surveillance techniques to map the distribution of mosquitoes. This effect is particularly severe in resource-constrained areas, as currently used methods like trapping and manual identification require considerable expense, labour, and time. However, for disease control strategies to be truly

Captura com celulares

Torres · 2025



Dengue-Infected Mosquito Detection with Uncertainty Evaluation of the Wingbeats using Monte Carlo Dropout

Israel Torres, Mariko Nakano, *Member, IEEE*, Jorge Cime-Castillo, Enrique Escamilla-Hernandez, *Member, IEEE*, Osvaldo Lopez-Garcia, and Humberto Lanz-Mendoza

***Abstract*— Considering that Aedes mosquitoes are a principal vector of the Dengue virus, which can cause the death of infected people in the worst case, accurate detection of infected Aedes mosquitoes is very important to prevent the further spread of the virus. This paper proposes a detection algorithm for infected *Aedes aegypti* mosquitoes by Dengue Virus-2 using their wingbeat signals. The proposed algorithm uses Long Short-Term Memory (LSTM) as a classifier of the input wingbeat signal into healthy mosquitoes and infected mosquitoes. All living beings, even those of the same species, have different characteristics depending on the season in which they are born, temperature, humidity, food, etc. Therefore, the wingbeat signals of mosquitoes are not always the same, although it is fed by the same blood infected. Considering these individual differences, an uncertainty measure based on Monte-Carlo dropout is introduced. The proposed algorithm detects approximately 5% of uncertainty data from all input wingbeat signals in the test set and provides a classification accuracy of 94.87% without any uncertainty.**

Link to graphical and video abstracts, and to code: https://latamt.ieeer9.org/index.php/transactions/article/view/9221

***Index Terms*—dengue fever, infected mosquitoes, Long Short-Term Memory, Monte Carlo dropout, uncertainty**

I. INTRODUCTION

CERTAIN mosquito species are responsible for the transmission of diseases such as dengue fever, Chikungunya fever, Zika, and malaria. Notably, *Aedes aegypti* and *Aedes albopictus* are the primary vectors of dengue fever, which can, in severe cases, result in mortality among infected individuals. According to a report from the Centers for Disease Control and Prevention (CDC) in the United States, approximately four billion people in the world are at risk of

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J. Cime-Castillo, and H. Lanz-Mendoza are with Instituto Nacional de Salud Publica, Cuernavaca, Morelos, Mexico (e-mails: jorge.cime@insp.mx, and hlanzm@insp.mx).

contracting this disease [1]. Many Latin American countries are located in tropical or subtropical regions, which coincide with the habitat of the *Aedes* mosquito. Therefore, these countries face the common problem of spreading the diseases transmitted by *Aedes* mosquitoes, especially dengue, which accounts for approximately 90% of all cases on the American continent [2].

A countermeasure to prevent further expansion of this disease is massive fumigation to exterminate these species of mosquitoes. However, massive fumigation causes irreversible ecological damage in addition to economic burden. Therefore, in many countries, fumigation of a specific area is only permitted when human dengue cases are reported. This is because detecting infected mosquitoes is only possible through molecular biology in specialized laboratories, which is time-consuming and expensive [3].

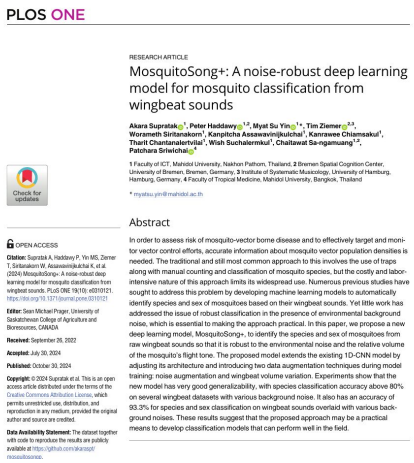
Considering the above-mentioned situations, the final objective of this paper is to propose a technological solution to detect dengue-infected *Aedes aegypti* mosquitoes in the field. The detection must be rapid to avoid a possible virus spread between humans, and detection accuracy must be as high as possible, with the lowest uncertainty, to perform effective fumigation. Effective fumigation minimizes the impact on ecosystems and reduces the economic burden. This research could potentially contribute to two of the United Nations Sustainable Development Goals (SDGs): Goal 3, "Good health and well-being," and Goal 15, "Life on Land."

Some scientists have reported that the behavior of infected mosquitoes has changed compared to that of healthy mosquitoes [4-6]. For instance, the authors of [4] and [5] reported that the infected female *Aedes* mosquitoes bite more frequently to obtain the same quantity of blood required by her eggs. The authors of [6] also observed that infected mosquitoes exhibited increased locomotor activity compared to healthy mosquitoes. These observations related to the change in behavior of infected mosquitoes suggest a possible hypothesis about the change in flight form of infected mosquitoes, and the possibility that infected mosquitoes can be detected by analyzing their wingbeat signals, which directly reflect the flight form of mosquitoes.

All living organisms, even those of the same species, have individual differences. The differences between individual mosquitoes lie in their birth season, feeding habits during their immature stages, and environmental factors such as temperature and humidity. These individual differences are

LSTM e incerteza

Supratak · 2024



PLOS ONE

RESEARCH ARTICLE

MosquitoSong+: A noise-robust deep learning model for mosquito classification from wingbeat sounds

Akara Supratak[1], Peter Haddawy[1,2], Myat Su Yin[1]*, Tim Ziemer[2,3], Worameth Siritanakorn[1], Kanpitcha Assawavinijkulchai[1], Kanrawee Chiamsakul[1], Thanit Chantanalertvilai[1], Wish Suchalermkul[1], Chaitawat Sa-ngamuang[1,2], Patchara Sriwichai[4]

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Check for updates

OPEN ACCESS

Citation: Supratak A, Haddawy P, Yin MS, Ziemer T, Siritanakorn W, Assawavinijkulchai K, et al. (2024) MosquitoSong+: A noise-robust deep learning model for mosquito classification from wingbeat sounds. PLoS ONE 19(10): e0310121. https://doi.org/10.1371/journal.pone.0310121

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Data Availability Statement: The dataset together with code to reproduce the results are publicly available at https://github.com/akarasupt/mosquitosongs.

Abstract

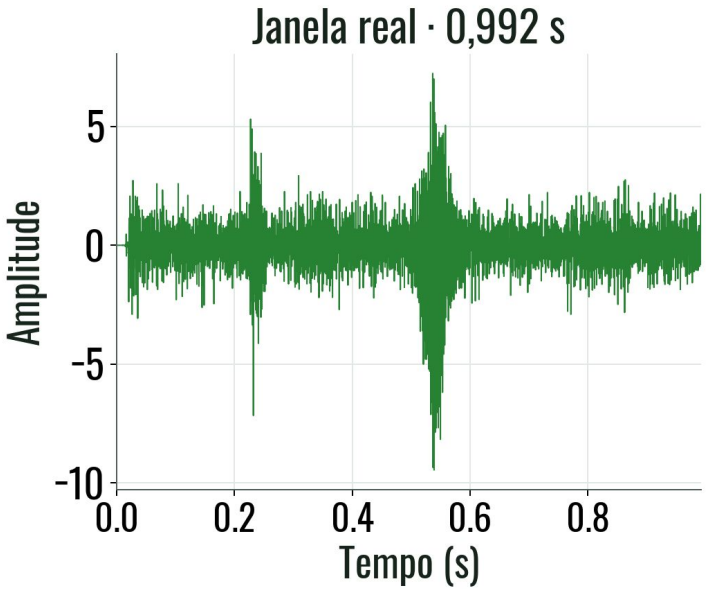
In order to assess risk of mosquito-vector borne disease and to effectively target and monitor vector control efforts, accurate information about mosquito vector population densities is needed. The traditional and still most common approach to this involves the use of traps along with manual counting and classification of mosquito species, but the costly and labor-intensive nature of this approach limits its widespread use. Numerous previous studies have sought to address this problem by developing machine learning models to automatically identify species and sex of mosquitoes based on their wingbeat sounds. Yet little work has addressed the issue of robust classification in the presence of environmental background noise, which is essential to making the approach practical. In this paper, we propose a new deep learning model, MosquitoSong+, to identify the species and sex of mosquitoes from raw wingbeat sounds so that it is robust to the environmental noise and the relative volume of the mosquito's flight tone. The proposed model extends the existing 1D-CNN model by adjusting its architecture and introducing two data augmentation techniques during model training: noise augmentation and wingbeat volume variation. Experiments show that the new model has very good generalizability, with species classification accuracy above 80% on several wingbeat datasets with various background noise. It also has an accuracy of 93.3% for species and sex classification on wingbeat sounds overlaid with various background noises. These results suggest that the proposed approach may be a practical means to develop classification models that can perform well in the field.

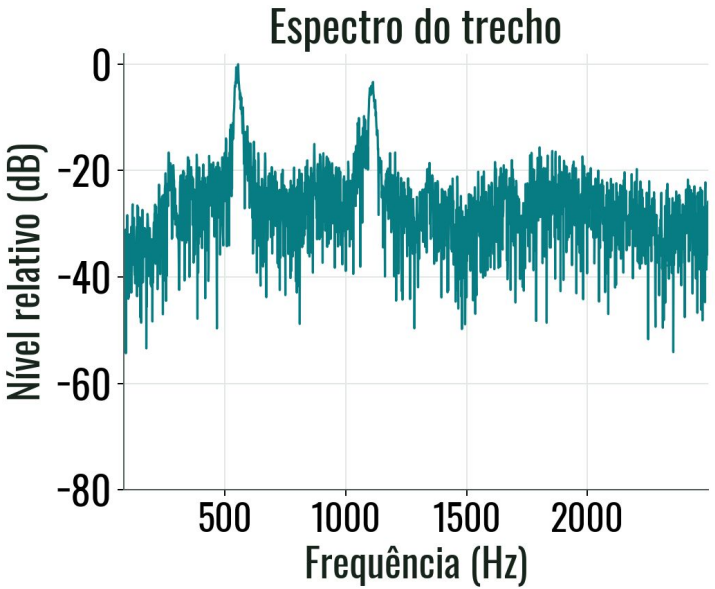
MosquitoSong+ · ruído

Da captura acústica à modelagem temporal e à robustez ao ruído.

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O sinal revela padrões no tempo e na frequência

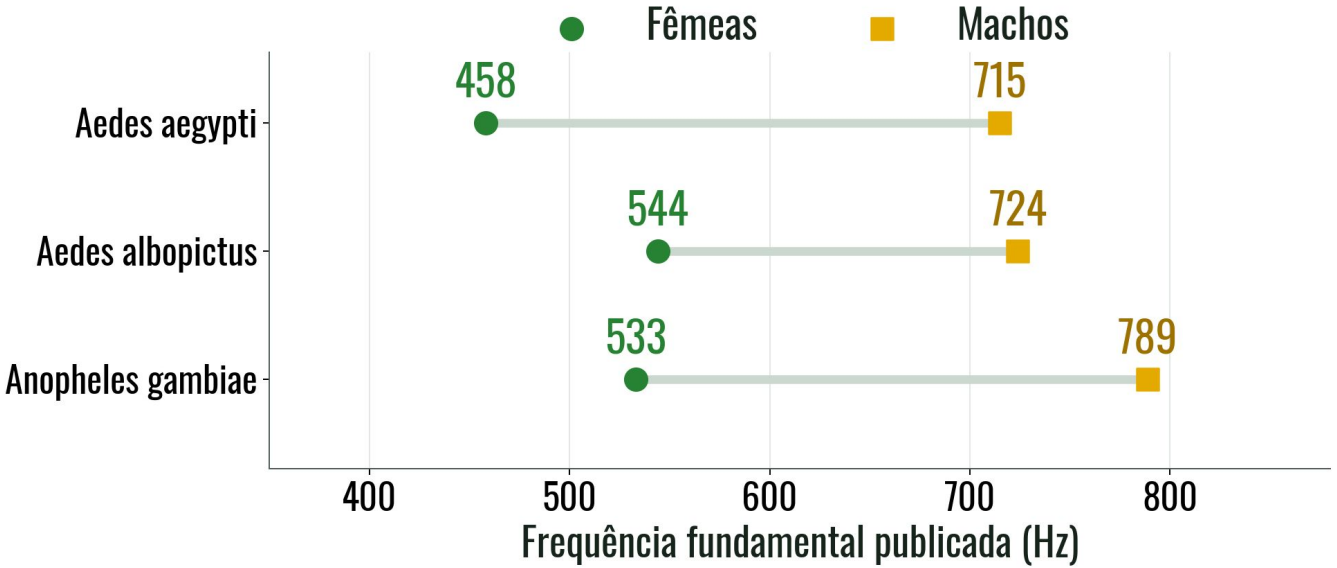




Trecho real de Aedes · 0,992 s · áudio a 8 kHz · espectro do mesmo trecho.

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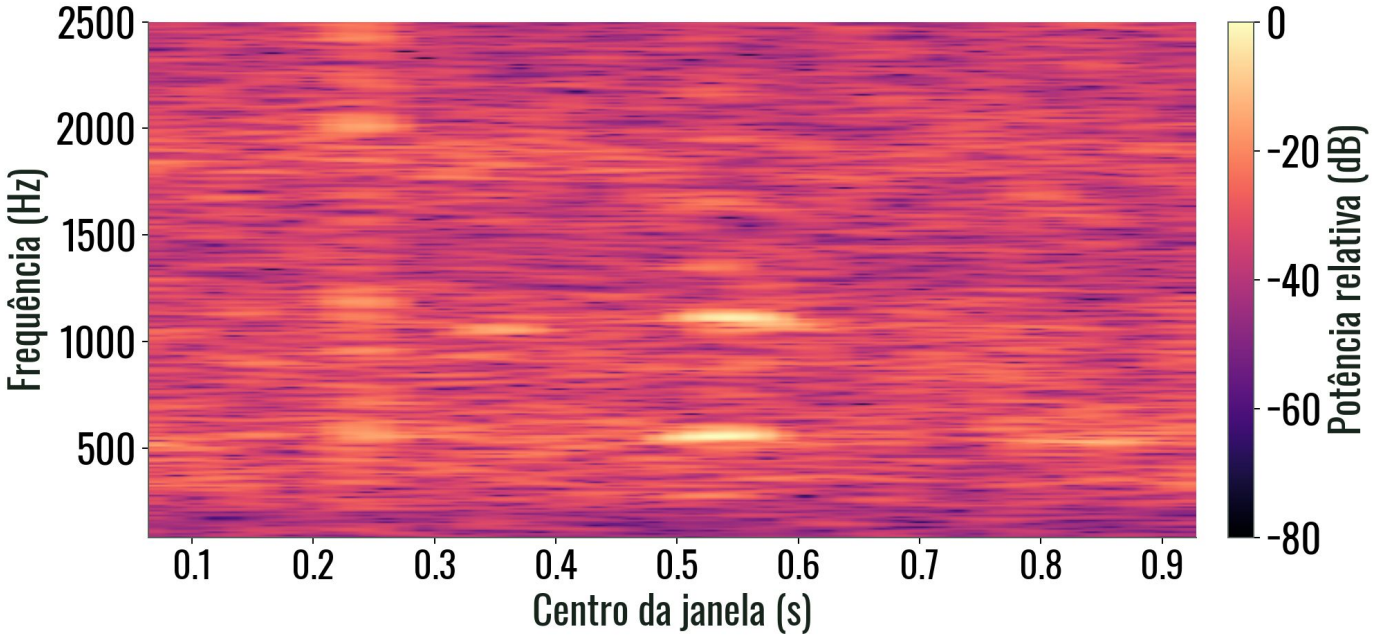
A frequência das asas ajuda a caracterizar o voo



Valores publicados a 25 °C · fêmeas e machos · Sinka et al. (2021).

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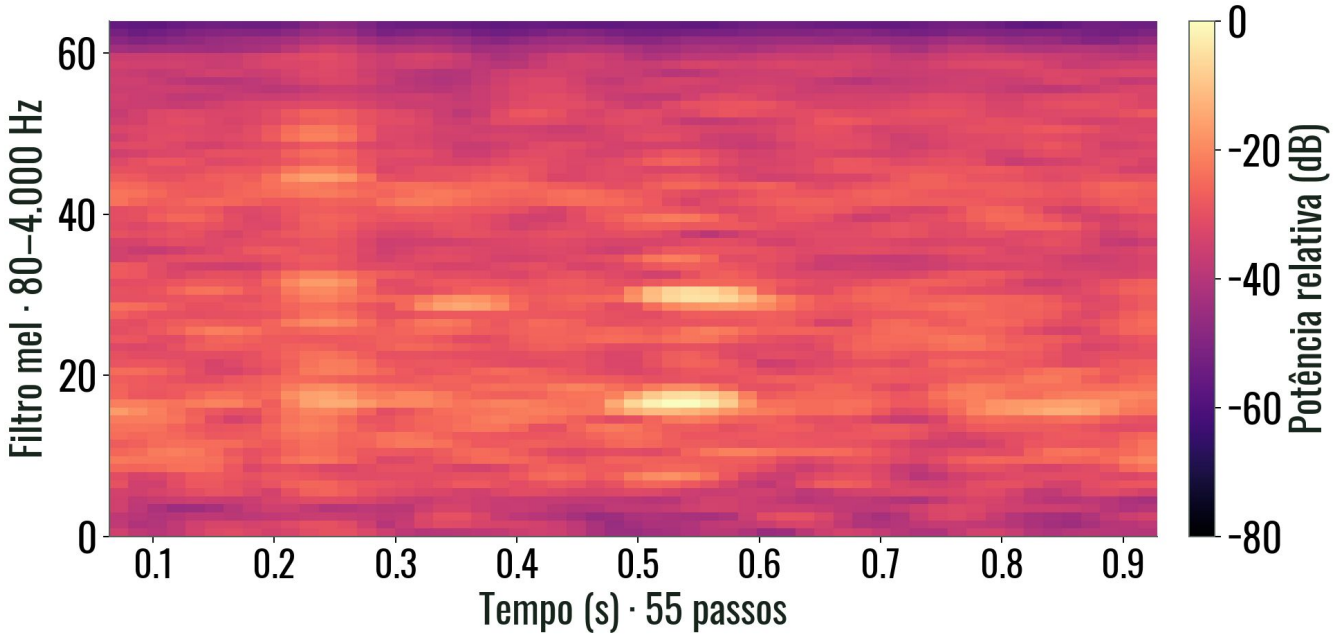
A STFT acompanha o espectro ao longo do voo



FFT 1.024 · Hann 128 ms · avanço 16 ms · separação entre bins: 7,8125 Hz.

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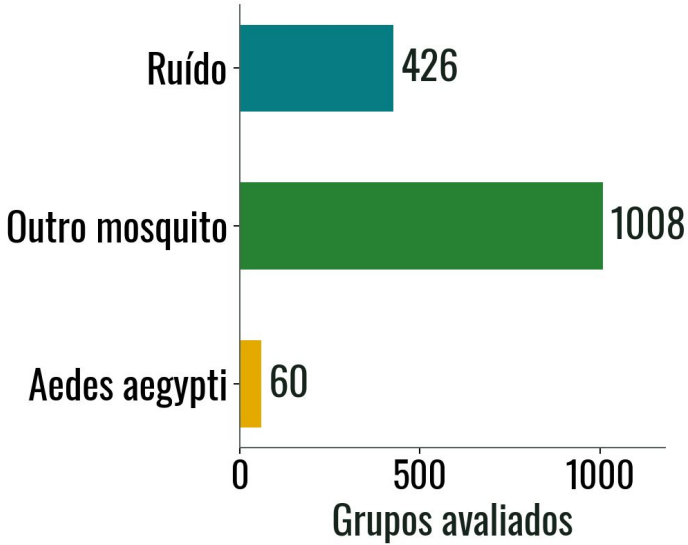
Do espectrograma à sequência mel

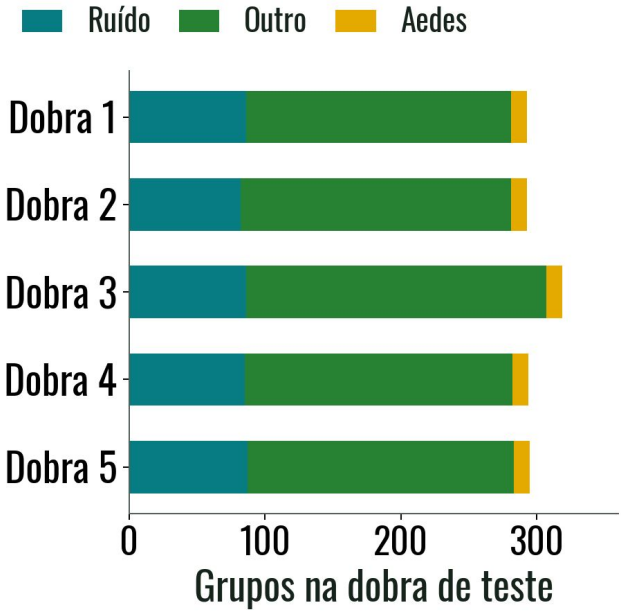


64 filtros · 55 passos · 80–4.000 Hz · normalização ajustada apenas no treino.

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Mesma população e cinco testes externos



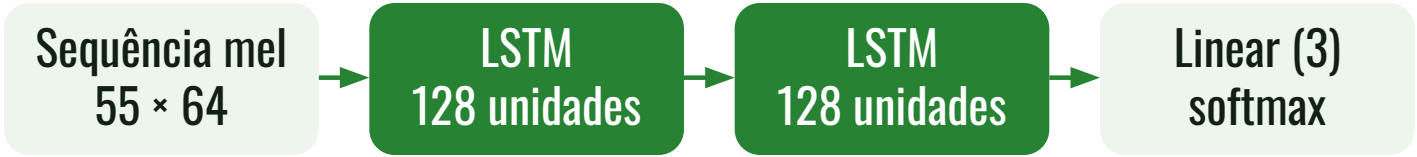


42.029 janelas · 1.494 grupos · 1.441 blocos de PCM · três classes.

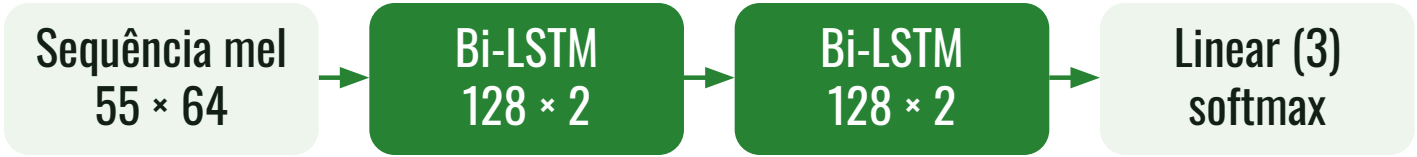
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Redes recorrentes para as três classes

LSTM · um sentido



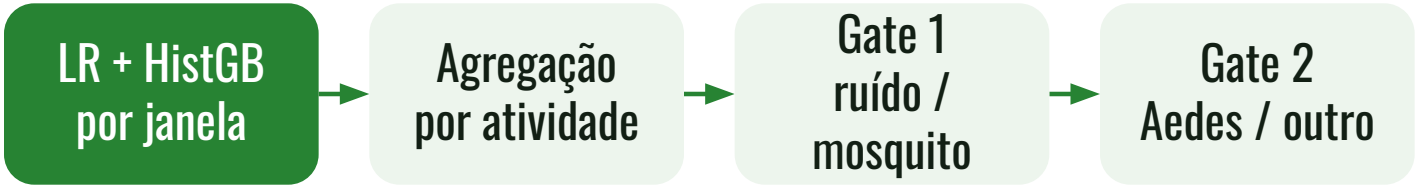
Bi-LSTM · dois sentidos



Duas camadas · 128 unidades por direção · dropout 0,20 · três saídas.

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A cascata clássica é a referência de comparação

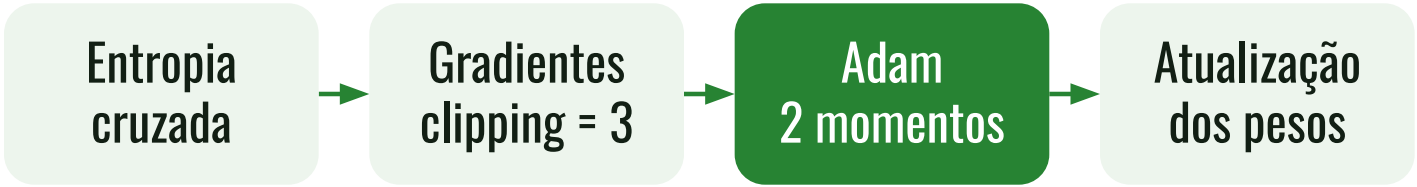


1.345 acertos em 1.494 grupos
90,03% de acurácia geral

LR + HistGB · probabilidades por janela · agregação por atividade · dois limiares.

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Adam ajusta os pesos das redes



Taxa de aprendizado: 0,0006
256 janelas por lote · até 30 épocas

Taxa 0,0006 · lotes de 256 · norma do gradiente limitada a 3.

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Regularização e seleção durante o treino

1 **Dropout entre camadas**

Probabilidade 0,20 nas duas arquiteturas.

2 **Amostragem por classe e gravação**

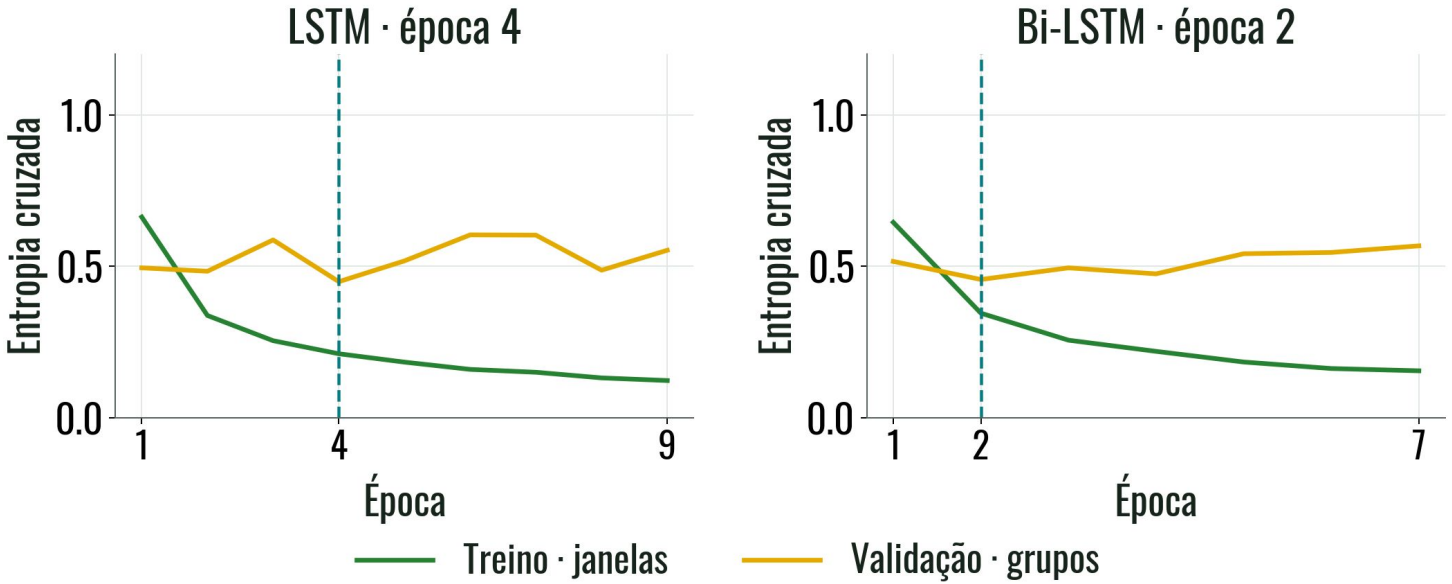
Gamma 0,50 · mesma massa para as três classes.

3 **Parada pela validação interna**

Menor perda por grupo · paciência de 5 épocas.

Monte Carlo dropout: 100 passagens na inferência de um novo áudio.

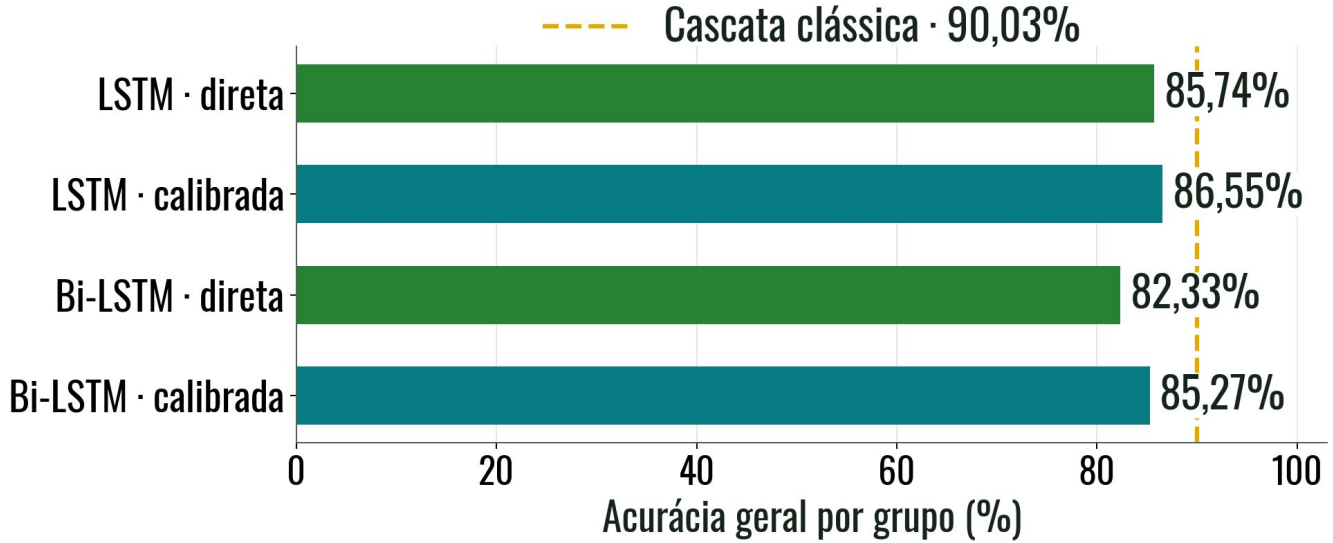
A validação escolhe a época de cada rede



Ajuste final · LSTM: época 4 · Bi-LSTM: época 2 · decisão pela validação interna.

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A cascata mantém 90,03% de acurácia geral



Mesmos 1.494 grupos · direta: maior probabilidade · calibrada: dois limiares internos.

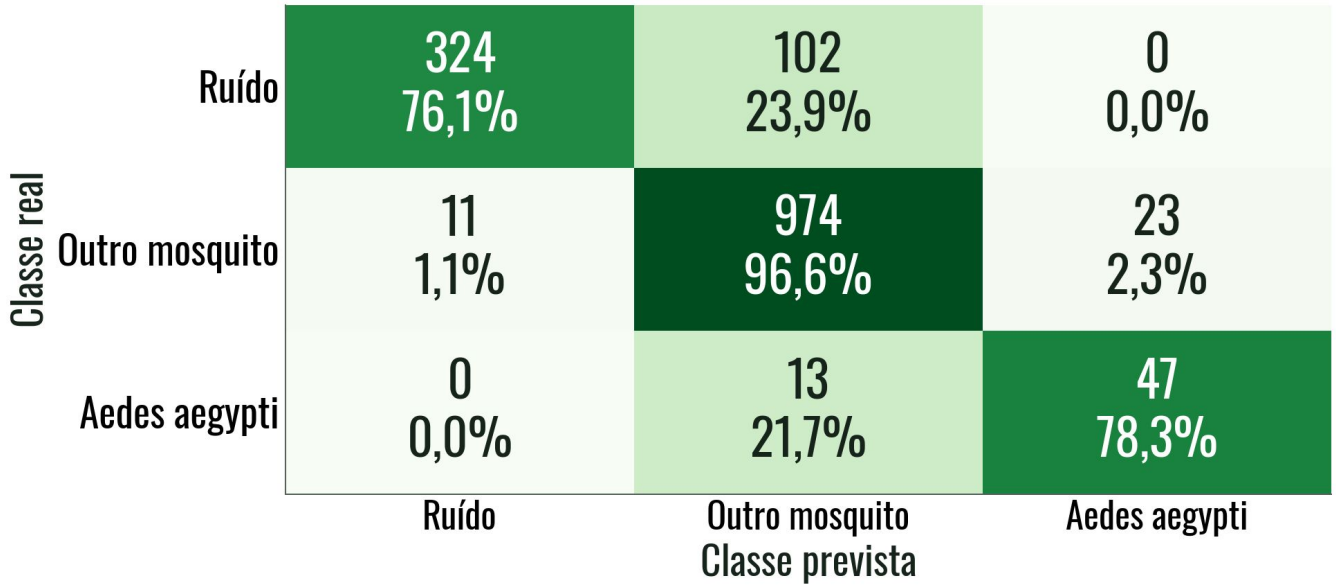
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Sensibilidade e falsos positivos mudam juntos

Modelo / regra	Sensibilidade Aedes	FPR ruído para Aedes	FPR outro para Aedes
Cascata clássica	78,33%	0,00%	2,28%
LSTM · direta	86,67%	3,76%	3,97%
LSTM · calibrada	33,33%	0,47%	0,60%
Bi-LSTM · direta	88,33%	4,93%	6,25%
Bi-LSTM · calibrada	28,33%	0,00%	0,40%

Sensibilidade: Aedes detectados · FPR: negativos classificados como Aedes.

Onde a cascata acerta e onde confunde

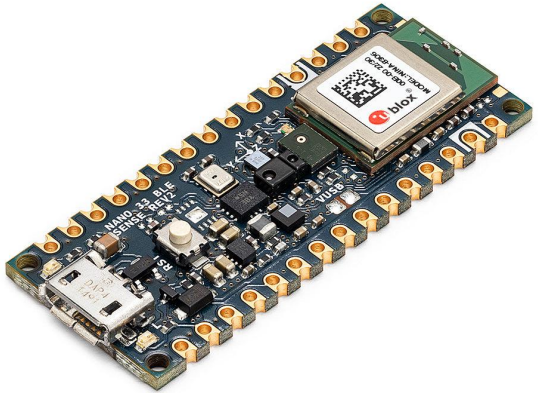


1.345/1.494 acertos · 47/60 Aedes detectados · linhas: real · colunas: previsão.

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Coleta futura: novo dataset com Arduino

Capturar o som
Microfone → USB → WAV
Anotar a coleta
Aedes / outros / ruído
Separar os dados
Por indivíduo e sessão



Nano 33 BLE Sense Rev2 · foto oficial Arduino

Base própria para treinar e validar o detector.

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Referências e créditos

Mukundarajan et al. (2017) · áudio e celulares · eLife

Torres et al. (2025) · LSTM, Bi-LSTM e MC dropout · IEEE

Supratak et al. (2024) · MosquitoSong+ · PLOS ONE

Sinka et al. (2021) · HumBug e frequências publicadas

Dados, código e resultados · github.com/Lciarallo/mosquito-wingbeat

Fotos: IFSC, CDC, Arduino, SES-RS e SESA-ES

Links completos, parâmetros e créditos nas notas de cada slide.

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Discussão final: limites e próximo ciclo

1 **Resultado e compromisso**

90,03% na cascata; redes diretas detectam mais Aedes.

2 **Avaliação retrospectiva**

Novas sessões e indivíduos ainda precisam ser avaliados.

3 **Próximo ciclo: robustez**

Congelar o detector e variar ruído, ambiente e microfone.

Comparação sem ablação causal · MosquitoSong separado · redes ainda sem porte embarcado.

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